

SECTION-B

Note: Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

- Q.11 Explain principle of operation of a steam condenser.
- Q.12 State & explain second law of thermodynamics in brief
- Q.13 Explain Otto cycle with a neat diagram.
- Q.14 Explain the basic concept of refrigerator.
- Q.15 Explain construction of fire tube boiler type steam generator.
- Q.16 Explain the working of reciprocating air compressor in detail.
- Q.17 Explain the concept & significance of entropy in thermodynamics.
- Q.18 The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is -20°C , determine the temperature of the surroundings to which it rejects heat.

SECTION-C

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

- Q.19 Explain the mechanism of four stroke petrol engine in detail.
- Q.20 Explain the principle & working of water tube boiler type steam generator in detail.

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Level-4, 2nd Sem/ DVOC(Ref. & Air Cond.) Subject : Basics of Thermodynamics

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Very Short answer type questions. All questions are compulsory (10x1=10)

- Q.1 Define compression ratio of an IC engine.
- Q.2 Give two examples of open system.
- Q.3 What are surroundings?
- Q.4 Define enthalpy.
- Q.5 What is a steady state flow process?
- Q.6 What are units of specific fuel consumption?
- Q.7 Define COP of heat pump.
- Q.8 Define closed system.
- Q.9 Define mean effective pressure.
- Q.10 A refrigerator has COP of 3. How much work must be supplied to the refrigerator in order to remove 200 J of heat from it?